

Test Chip for Bump Bond Yield Evaluation in High Density Flip Chip Technologies

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Abstract—Modern flip chip technologies for imaging applications have achieved a very high integration level together with the possibility of large assemblies. These developments have resulted in an enormous increase in the total number of bump bonds per assembly. As a result, yield tests become more difficult, and an accurate measurement of it is often discarded. This problem is aggravated in medical applications, as the critical information can be limited to a few pixels, and therefore, yields should be very close to 100%. In these cases, a difference of a few tenths percent in bump bond yield can make the difference between a usable and a non-usable assembly. For this reasons, quantitative and precise measurements of bump bond yield are needed to characterize the quality of any high density flip chip technology. In this contribution, we present a newly developed test chip for quantitative electrical characterization of the bump bond yield of high density flip chip technologies, allowing optimization and statistical control of the process. This test chip facilitates the identification of possible technological problems with precise quantitative yield measurements. It also allows to pinpoint any localized systematic failure in the bump-bonding process. The test chip has been used to evaluate the yield of different flip chip technologies and has contributed to their fine optimization.

I. INTRODUCTION AND RELEVANCE

MODERN detector applications, for X-ray and gamma ray imaging, make use of high density pixel arrays that should be connected with their corresponding readout electronics via bump bonding or flip chip technologies. The technology developments achieved lately have increased the density of the pixel arrays along with the size of the assemblies. These developments have resulted in an enormous increase in the total number of bump bonds per assembly [1]. As this number increases, quality and yield test of the flip chip technologies becomes more difficult, and an accurate control of these parameters is often discarded. Test structures and full test chips have been used [2][3], but mostly for bond quality

tests or partial yield estimations. X-ray or Scanning Acoustic Microscopy (CSAM) are also applicable [4], but they can overpass some bond problems and they are impractical for quantitative measurements of bump bond yield for large assemblies. This problem is aggravated in applications like medical imaging, as the critical information can be limited to a few pixels, and therefore yields should be very close to 100%. In these cases, a difference of a few tenths percent in bump bond yield can make the difference between a usable and a non-usable assembly. For this reasons precise and quantitative measurements of bump bond yield are needed to characterize the quality of modern flip chip technologies.

In this paper we present a newly developed test structure that allows quantitative electrical characterization of the bump bond yield of a technology, allowing statistical control of the process. The test structure consists of a set of bump bond chains evenly distributed across the assembly. By measuring the conductivity of these chains, the bump bond yield of the whole assembly can be calculated statistically. The number of bump bonds per chain can be tuned in order to adjust the test to the yield ranges of interest for the application. This test chip has been used on different flip chip technologies in order to characterize their yield.

II. TEST STRUCTURE DESIGN

The test structure that has been designed for quantitative yield control of flip chip technologies consists of a series of bump bond daisy chains evenly distributed across the assembly. Each chain contains L bump bonds, connected in a way that if any of them fails, the whole chain is electrically open. In the particular case of the test chip used for the measurements described below, assemblies of 256x256 bump bonds with 55 μm pitch were tested. Each chain has 50 bump bonds and each assembly has 504 independent chains. In this way, two fifths of the bonds of the assembly are connected to a chain, and the rest are short-circuited. The chains are then connected to an array of test pads prepared for automatic measurements with an automatic probe table using a 1x16 probe card. A detail of the central ground bus area of the detector side of the test assembly can be seen in Fig. 1, while Fig. 2 shows an area of the probe card test pads in the chips side of the test assembly.

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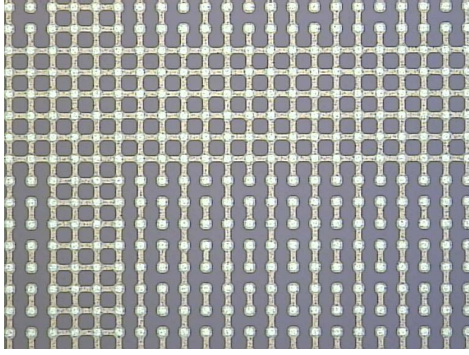


Fig. 1. Detail of the central ground bus area in the detector side area of the test structure.

Electrical conductivity measurements are performed on each of the test pads connected to each of the chains in the test assembly. The number of good chains will lead, by use of the Poisson distribution, to the calculation of the mean number of good bonds per chain, and then to the probability of good bonds per assembly or bump bond yield. The selection of the number of bonds per chain allows the optimization of the range of yields in which the test structure is more precise. In the case of the test structure designed for this paper, the number of bonds per chain is 50, which gives a better precision in the range of 95% to 100%. This is a range of bump bond yields of interest in the current technologies and where it falls the difference between a usable and a non-usable assembly.

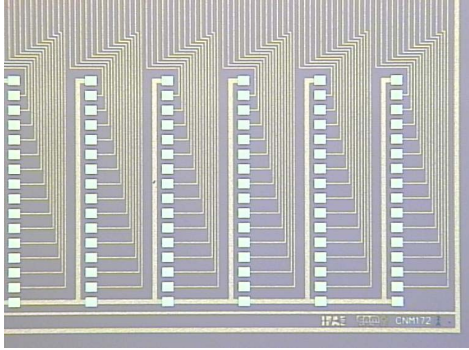


Fig. 2. detail of the probe card test pad array in the chip side.

II. YIELD CALCULATION

If we assume that the number of bad bonds is small with respect to the total number of bonds, then the distribution of bad bonds will follow the Poisson distribution. According to that, the probability of x bad bonds per chain can be written as:

$$P(x) = \frac{\mu^x}{x!} e^{-\mu} \quad (1)$$

where μ is the mean number of bad bonds per chain.

Let us now imagine that we perform an analysis on a test assembly with N total chains and L bonds per chain, and that

we obtain M “bad” (open circuit) chains of the total number. In this case, the probability of having a bad chain will be,

$$\frac{M}{N} = 1 - P(0) \quad (2)$$

because a “bad” chain is one that has any number of bad bonds different than zero, by design. Hence, according to (1)

$$\frac{M}{N} = 1 - e^{-\mu} \quad (3)$$

Then, if we consider that the mean number of bond per chain can be written in terms of the probability of a single bond to be bad (λ) as $\mu = \lambda \cdot L$ then we can write for λ according to (3)

$$\lambda = -\frac{1}{L} \cdot \log\left(1 - \frac{M}{N}\right) \quad (4)$$

Finally, if we define $\Lambda = M / N$ as the ratio of bad chains per assembly, then according to (4) the total bump bond yield of the assembly (Y) can be written in terms of Λ as

$$Y(\Lambda) = \frac{1}{L} \cdot \log(1 - \Lambda) + 1 \quad (5)$$

Fig. 2 shows the function $Y(\Lambda)$ for different values of the number of bonds per chain, L . As can be seen in that Figure, the number of bonds per chain can be tuned in order to adjust the test structure to the range of bump bond yields of more interest for the application.

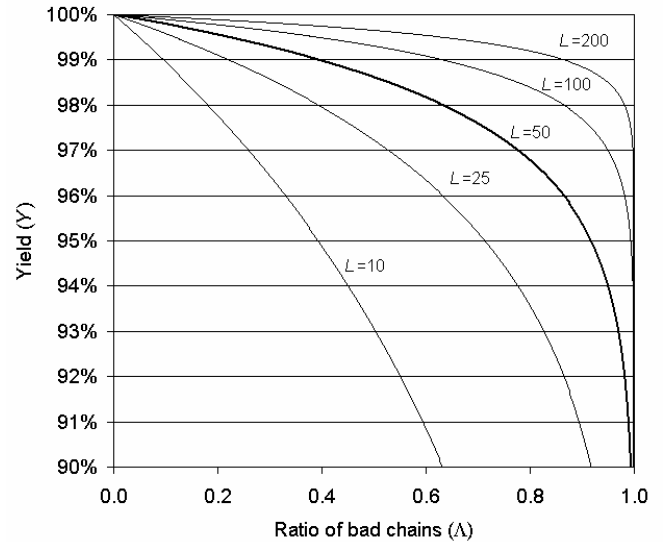


Fig. 2. Bump bond yield versus ratio of bad chains for different chain lengths.

III. TEST RESULTS

The test chip has been used to test the bump bond yield on different flip chip technologies in order to validate it. The test structures were fabricated on silicon in the clean room of the

Centro Nacional de Microelectrónica (CNM-CSIC) and then sent to the different flip chip manufacturers for the bonding. Then they were sent back for the test.

Table I shows the yield results obtained for the different assemblies measured from two distinct manufacturers. It can be seen from these results that assemblies from manufacturer A do not show a proper yield. With these data this manufacturer is tuning the process in order to obtain higher yields. This reveals that the test structure presented here facilitates the identification of possible technological problems with precise quantitative measurements. It is important to note that differentiation among yields in the range of 95% to 100% when the bad bonds are uniformly distributed across the assembly is not straightforward, but it is critical for imaging applications. This test structure is a valid tool for these cases as can be seen from the measurements.

TABLE I
YIELD RESULTS FOR DIFFERENT ASSEMBLIES AND MANUFACTURERS

		Bad Chain Ratio (Δ)	YIELD (Y)
Manufacturer A	Assembly #1	0.946	94.15%
	Assembly #2	0.980	92.16%
	Assembly #3	0.762	97.13%
	Assembly #4	0.990	90.77%
Manufacturer B	Assembly #5	0.052	99.89%

Another feature of this test structure that is worthy of note is that it also helps to pin point any localized systematic failure in the bump-bonding process. As the position of each daisy chain in the assembly is well known by the designer, and each daisy chain is connected also to a well known test pad, a representation of the results in terms of the geometrical position of each chain is straight forward. This diagram will allow an easy identification of any systematic technological failure in the bump bond process.

IV. CONCLUSION

With the test chip presented here one can precisely evaluate the bump-bonding yield and use this result to characterize and optimize the process. Although this structure does not cover the effects of the bump-bonding process on the performance of detectors itself, it is a very useful tool for manufacturers in the area of medical imaging applications. By the precise information on bump bond yield that it provides, manufacturers can have a better insight on the performance and limits of their final detector system.

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